WNBA 2028 Player Projection Model V2

Your Team Names

April 2026

Libraries

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(broom)
library(knitr)
library(kableExtra)
```

Step 1: Load and Clean Merged Dataset

```
master <- read.csv('/Users/shashwatmishra/Downloads/all_years_merged.csv') %>%
  rename(season      = Season,
         player_name = Name,
         team        = TeamAbbreviation) %>%
  mutate(
    ppg    = Points    / GamesPlayed,
    rpg    = Rebounds  / GamesPlayed,
    apg    = Assists   / GamesPlayed,
    bpg    = Blocks    / GamesPlayed,
    spg    = Steals    / GamesPlayed,
    tpg    = Turnovers / GamesPlayed,
    min_pg = Minutes   / GamesPlayed
  ) %>%
  filter(Minutes >= 100)

cat("Total rows:", nrow(master), "\n")
```

```
## Total rows: 691
```

```
cat("Seasons:", paste(sort(unique(master$season)), collapse = ", "), "\n")
```

```
## Seasons: 2021, 2022, 2023, 2024, 2025
```

```
cat("Unique players:", n_distinct(master$player_name), "\n")
```

```
## Unique players: 260
```

Step 2: Add Age

```
players_2025 <- read.csv('/Users/shashwatmishra/Downloads/WNBA Players 2025 - Sheet1.csv') %>%
  select(player_name = PLAYER, age_2025 = AGE)

master <- master %>%
  left_join(players_2025, by = "player_name") %>%
  mutate(age = age_2025 - (2025 - season)) %>%
  filter(!is.na(age))

cat("Rows after age join:", nrow(master), "\n")
```

```
## Rows after age join: 444
```

```
cat("Players retained:", n_distinct(master$player_name), "\n")
```

```
## Players retained: 131
```

Step 3: Create Lagged Features

```
model_data <- master %>%
  arrange(player_name, season) %>%
  group_by(player_name) %>%
  mutate(
    lag1_ortg = lag(OnOffRtg, 1),
    lag1_drtg = lag(OnDefRtg, 1),
    lag1_ppg  = lag(ppg,      1),
    lag1_rpg  = lag(rpg,      1),
    lag1_apg  = lag(apg,      1),
    lag1_ts   = lag(TsPct,    1),
    lag1_usg  = lag(Usage,    1),
    lag1_efg  = lag(EfgPct,   1),
    lag1_tpg  = lag(tpg,      1),
    lag1_spg  = lag(spg,      1),
    lag1_bpg  = lag(bpg,      1)
  ) %>%
  ungroup() %>%
  filter(!is.na(lag1_ortg))

cat("Model dataset rows:", nrow(model_data), "\n")
```

```
## Model dataset rows: 313
```

```
cat("Seasons covered:", paste(sort(unique(model_data$season)), collapse = ", "), "\n")
```

```
## Seasons covered: 2022, 2023, 2024, 2025
```

```r
cat("Unique players:", n_distinct(model_data$player_name), "\n")
```

```
## Unique players: 105
```

Step 4: Train/Test Split (80/20)

```r
set.seed(42)
train_idx <- sample(nrow(model_data), 0.8 * nrow(model_data))
train_set <- model_data[ train_idx, ]
test_set  <- model_data[-train_idx, ]

cat("Training rows:", nrow(train_set), "\n")
```

```
## Training rows: 250
```

```r
cat("Test rows:", nrow(test_set), "\n")
```

```
## Test rows: 63
```

Step 5: ORTG and DRTG Models

```r
library(car)
ortg_model <- lm(OnOffRtg ~ lag1_ortg + lag1_ppg + lag1_ts +
                              lag1_usg + lag1_apg + lag1_tpg + age,
                 data = train_set)

drtg_model <- lm(OnDefRtg ~ lag1_drtg + lag1_rpg + lag1_spg +
                              lag1_bpg + lag1_tpg + age,
                 data = train_set)

summary(ortg_model)
```

```
##
## Call:
## lm(formula = OnOffRtg ~ lag1_ortg + lag1_ppg + lag1_ts + lag1_usg +
##     lag1_apg + lag1_tpg + age, data = train_set)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.2894  -3.1943   0.0857   3.0603  11.1798
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 81.12009    7.66255  10.587  < 2e-16 ***
## lag1_ortg    0.20179    0.06666   3.027 0.002737 **
## lag1_ppg     0.45090    0.13354   3.377 0.000855 ***
## lag1_ts      4.45512    6.25623   0.712 0.477083
```

```
## lag1_usg    -0.20892    0.13053  -1.601 0.110770
## lag1_apg     0.72708    0.32049   2.269 0.024170 *
## lag1_tpg    -1.08439    0.77906  -1.392 0.165225
## age         -0.04141    0.09016  -0.459 0.646412
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.765 on 242 degrees of freedom
## Multiple R-squared:  0.2792, Adjusted R-squared:  0.2583
## F-statistic: 13.39 on 7 and 242 DF,  p-value: 1.368e-14
```

```r
summary(drtg_model)
```

```
##
## Call:
## lm(formula = OnDefRtg ~ lag1_drtg + lag1_rpg + lag1_spg + lag1_bpg +
##     lag1_tpg + age, data = train_set)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.8356  -3.2072  -0.3437   3.2426  12.4756
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 83.58111    7.05055  11.855  < 2e-16 ***
## lag1_drtg    0.21555    0.06045   3.566 0.000437 ***
## lag1_rpg     0.23808    0.18806   1.266 0.206733
## lag1_spg    -2.16176    0.83968  -2.575 0.010632 *
## lag1_bpg    -2.12355    0.88948  -2.387 0.017733 *
## lag1_tpg     0.42182    0.51397   0.821 0.412619
## age         -0.04359    0.08594  -0.507 0.612508
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.561 on 243 degrees of freedom
## Multiple R-squared:  0.1345, Adjusted R-squared:  0.1132
## F-statistic: 6.296 on 6 and 243 DF,  p-value: 3.612e-06
```

```r
vif(ortg_model)
```

```
## lag1_ortg  lag1_ppg   lag1_ts  lag1_usg  lag1_apg  lag1_tpg       age
##  1.523041  6.193167  1.613866  4.436801  2.940676  3.928065  1.070379
```

```r
vif(drtg_model)
```

```
## lag1_drtg  lag1_rpg  lag1_spg  lag1_bpg  lag1_tpg       age
##  1.110691  2.390826  1.782971  1.844894  1.866093  1.061575
```

Step 5b: Trimmed Models (Significant Predictors Only)

```r
ortg_model_v2 <- lm(OnOffRtg ~ lag1_ortg + lag1_ppg + lag1_apg,
                    data = train_set)

drtg_model_v2 <- lm(OnDefRtg ~ lag1_drtg + lag1_spg + lag1_bpg,
                    data = train_set)

summary(ortg_model_v2)
```

```
##
## Call:
## lm(formula = OnOffRtg ~ lag1_ortg + lag1_ppg + lag1_apg, data = train_set)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -19.558  -3.309  -0.223   2.925  12.981
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 71.14116    5.89098  12.076  < 2e-16 ***
## lag1_ortg    0.28243    0.05957   4.741  3.6e-06 ***
## lag1_ppg     0.22745    0.06616   3.438 0.000688 ***
## lag1_apg     0.43676    0.22276   1.961 0.051038 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.803 on 246 degrees of freedom
## Multiple R-squared:  0.2553, Adjusted R-squared:  0.2463
## F-statistic: 28.12 on 3 and 246 DF,  p-value: 1.145e-15
```

```r
summary(drtg_model_v2)
```

```
##
## Call:
## lm(formula = OnDefRtg ~ lag1_drtg + lag1_spg + lag1_bpg, data = train_set)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.565  -3.287  -0.539   3.054  13.410
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 81.69632    6.20358  13.169  < 2e-16 ***
## lag1_drtg    0.22764    0.05871   3.877 0.000136 ***
## lag1_spg    -1.36889    0.66506  -2.058 0.040615 *
## lag1_bpg    -1.46498    0.68691  -2.133 0.033939 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.569 on 246 degrees of freedom
## Multiple R-squared:  0.1207, Adjusted R-squared:   0.11
## F-statistic: 11.26 on 3 and 246 DF,  p-value: 6.025e-07
```

Step 5c: Final Model With Efficiency Check

```r
ortg_model_v3 <- lm(OnOffRtg ~ lag1_ortg + lag1_ppg +
                                lag1_apg + lag1_ts,
                    data = train_set)

drtg_model_v3 <- lm(OnDefRtg ~ lag1_drtg + lag1_spg + lag1_bpg,
                    data = train_set)

summary(ortg_model_v3)
```

```
##
## Call:
## lm(formula = OnOffRtg ~ lag1_ortg + lag1_ppg + lag1_apg + lag1_ts,
##     data = train_set)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.2699  -3.3539  -0.0354   2.9477  11.9072
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 70.28501    5.91487  11.883  < 2e-16 ***
## lag1_ortg    0.25151    0.06370   3.948 0.000103 ***
## lag1_ppg     0.19372    0.07058   2.745 0.006508 **
## lag1_apg     0.49521    0.22652   2.186 0.029754 *
## lag1_ts      8.00986    5.91267   1.355 0.176763
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.795 on 245 degrees of freedom
## Multiple R-squared:  0.2609, Adjusted R-squared:  0.2488
## F-statistic: 21.62 on 4 and 245 DF,  p-value: 2.731e-15
```

```r
summary(drtg_model_v3)
```

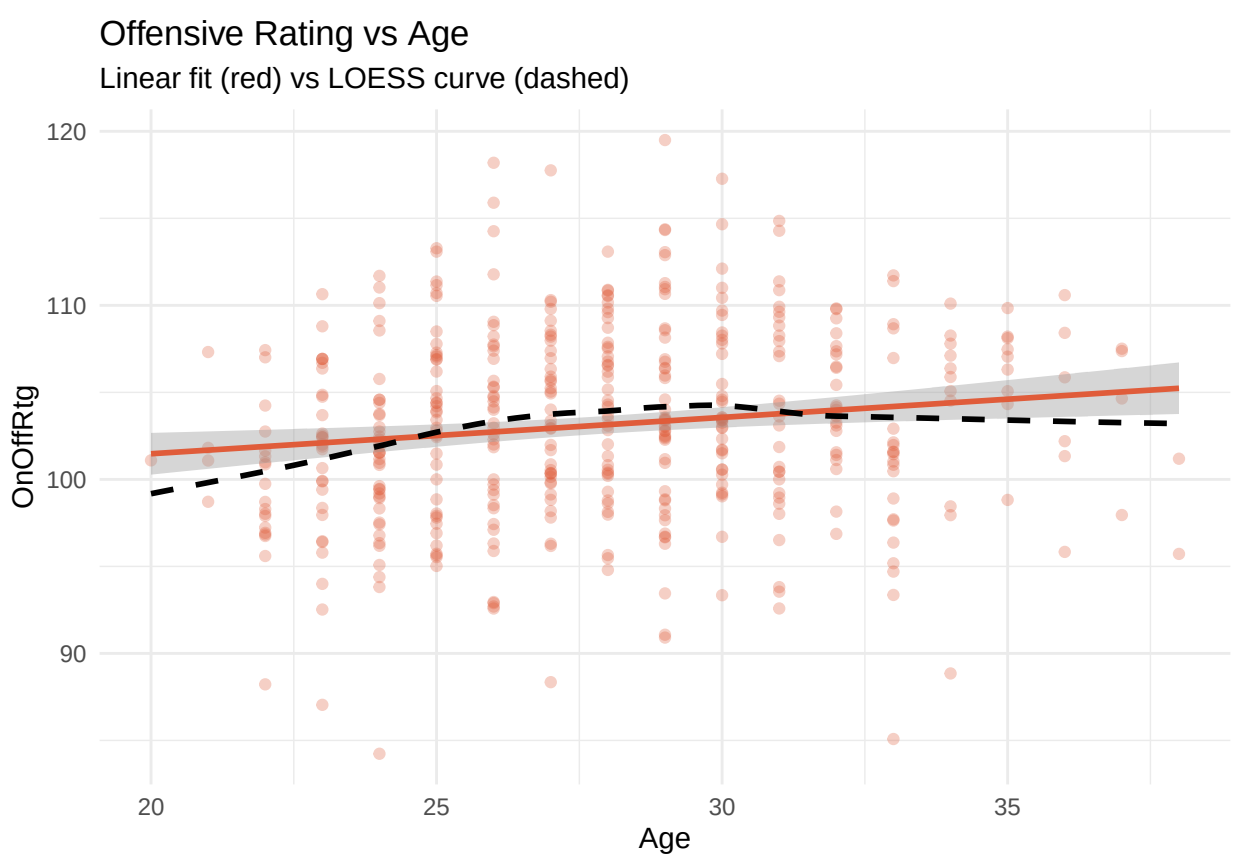
```
##
## Call:
## lm(formula = OnDefRtg ~ lag1_drtg + lag1_spg + lag1_bpg, data = train_set)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.565  -3.287  -0.539   3.054  13.410
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 81.69632    6.20358  13.169  < 2e-16 ***
## lag1_drtg    0.22764    0.05871   3.877 0.000136 ***
## lag1_spg    -1.36889    0.66506  -2.058 0.040615 *
## lag1_bpg    -1.46498    0.68691  -2.133 0.033939 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 4.569 on 246 degrees of freedom
## Multiple R-squared:  0.1207, Adjusted R-squared:   0.11
## F-statistic: 11.26 on 3 and 246 DF,  p-value: 6.025e-07
```

Step 6: Age Analysis

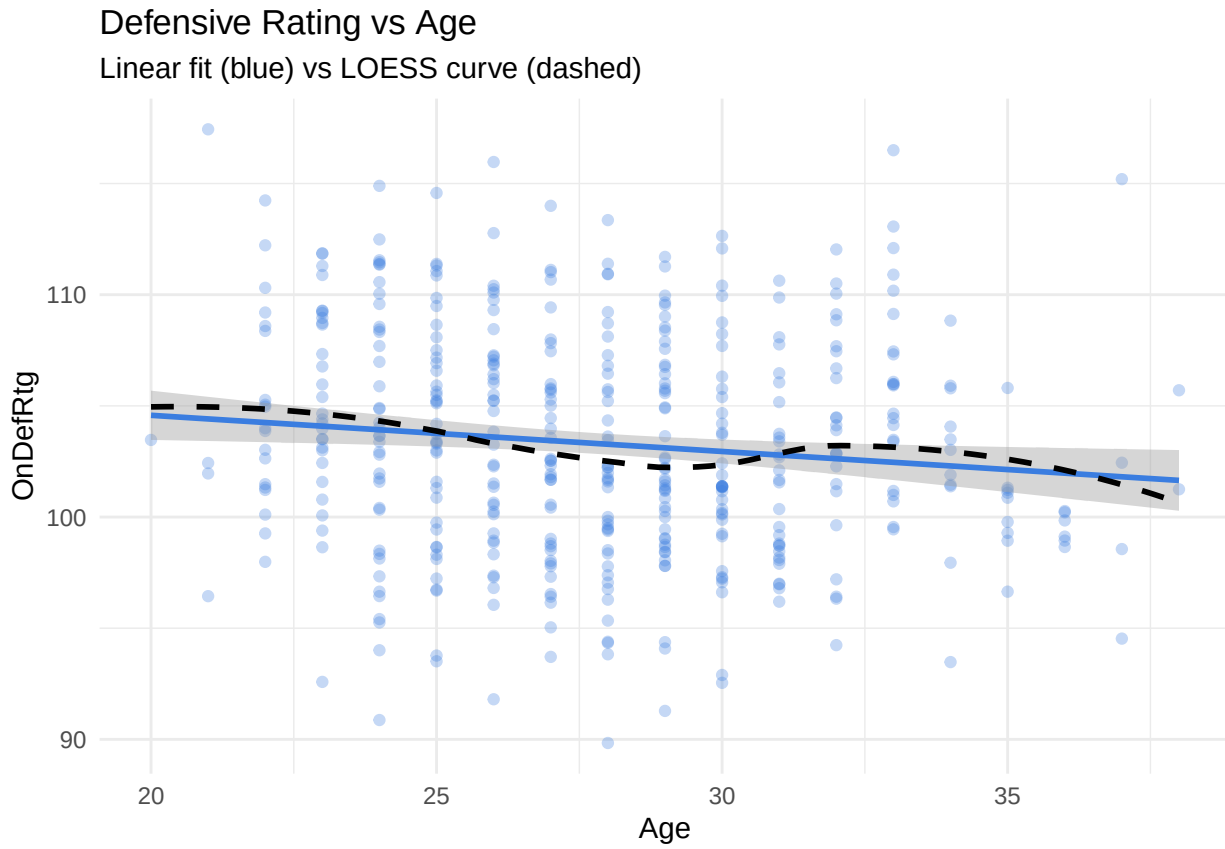
```r
library(ggplot2)

# ORTG vs age
ggplot(master, aes(x = age, y = OnOffRtg)) +
  geom_point(alpha = 0.3, color = "#e05c3a") +
  geom_smooth(method = "lm", color = "#e05c3a", se = TRUE) +
  geom_smooth(method = "loess", color = "black", linetype = "dashed", se = FALSE) +
  labs(title = "Offensive Rating vs Age",
       subtitle = "Linear fit (red) vs LOESS curve (dashed)",
       x = "Age", y = "OnOffRtg") +
  theme_minimal()
```




```r
# DRTG vs age
ggplot(master, aes(x = age, y = OnDefRtg)) +
  geom_point(alpha = 0.3, color = "#3a7de0") +
  geom_smooth(method = "lm", color = "#3a7de0", se = TRUE) +
  geom_smooth(method = "loess", color = "black", linetype = "dashed", se = FALSE) +
```

```r
  labs(title = "Defensive Rating vs Age",
       subtitle = "Linear fit (blue) vs LOESS curve (dashed)",
       x = "Age", y = "OnDefRtg") +
  theme_minimal()
```




```r
# Also check average rating by age bucket
master %>%
  mutate(age_bucket = cut(age, breaks = c(20, 24, 27, 30, 33, 36, 45),
                          labels = c("21-24", "25-27", "28-30",
                                     "31-33", "34-36", "37+"))) %>%
  group_by(age_bucket) %>%
  summarise(
    n            = n(),
    avg_ortg     = round(mean(OnOffRtg, na.rm = TRUE), 1),
    avg_drtg     = round(mean(OnDefRtg, na.rm = TRUE), 1),
    avg_netrtg   = round(mean(OnOffRtg - OnDefRtg, na.rm = TRUE), 1)
  ) %>%
  print()
```

```
## # A tibble: 7 x 5
##   age_bucket     n avg_ortg avg_drtg avg_netrtg
##   <fct>      <int>    <dbl>    <dbl>      <dbl>
## 1 21-24         91     101.     104.       -3.6
## 2 25-27        119     103.     104.       -0.1
## 3 28-30        126     104.     102.        1.7
```

```
## 4 31-33         75     103.     104.       -0.5
## 5 34-36         26     105.     101.        3.5
## 6 37+            6     102.     103.       -0.5
## 7 <NA>           1     101.     104.       -2.4
```

Step 7a: Test Set Evaluation

```r
test_set$pred_ortg <- predict(ortg_model_v3, newdata = test_set)
test_set$pred_drtg <- predict(drtg_model_v3, newdata = test_set)

cat("ORTG Train MSE:", round(mean(ortg_model_v3$residuals^2), 4), "\n")
```

```
## ORTG Train MSE: 22.5348
```

```r
cat("ORTG Test MSE: ", round(mean((test_set$OnOffRtg - test_set$pred_ortg)^2, na.rm=TRUE), 4), "\n")
```

```
## ORTG Test MSE:  20.1892
```

```r
cat("DRTG Train MSE:", round(mean(drtg_model_v3$residuals^2), 4), "\n")
```

```
## DRTG Train MSE: 20.5409
```

```r
cat("DRTG Test MSE: ", round(mean((test_set$OnDefRtg - test_set$pred_drtg)^2, na.rm=TRUE), 4), "\n")
```

```
## DRTG Test MSE:  24.2465
```

Step 8: Project 2028

```r
proj_2025 <- master %>%
  filter(season == 2025, Minutes >= 300) %>%
  group_by(player_name) %>%
  slice_max(order_by = Minutes, n = 1) %>%
  ungroup() %>%
  rename(
    lag1_ortg = OnOffRtg,
    lag1_drtg = OnDefRtg,
    lag1_ppg  = ppg,
    lag1_apg  = apg,
    lag1_ts   = TsPct,
    lag1_spg  = spg,
    lag1_bpg  = bpg
  ) %>%
  mutate(age = age + 3)

proj_2025$proj_ortg_2028 <- predict(ortg_model_v3, newdata = proj_2025)
proj_2025$proj_drtg_2028 <- predict(drtg_model_v3, newdata = proj_2025)
```

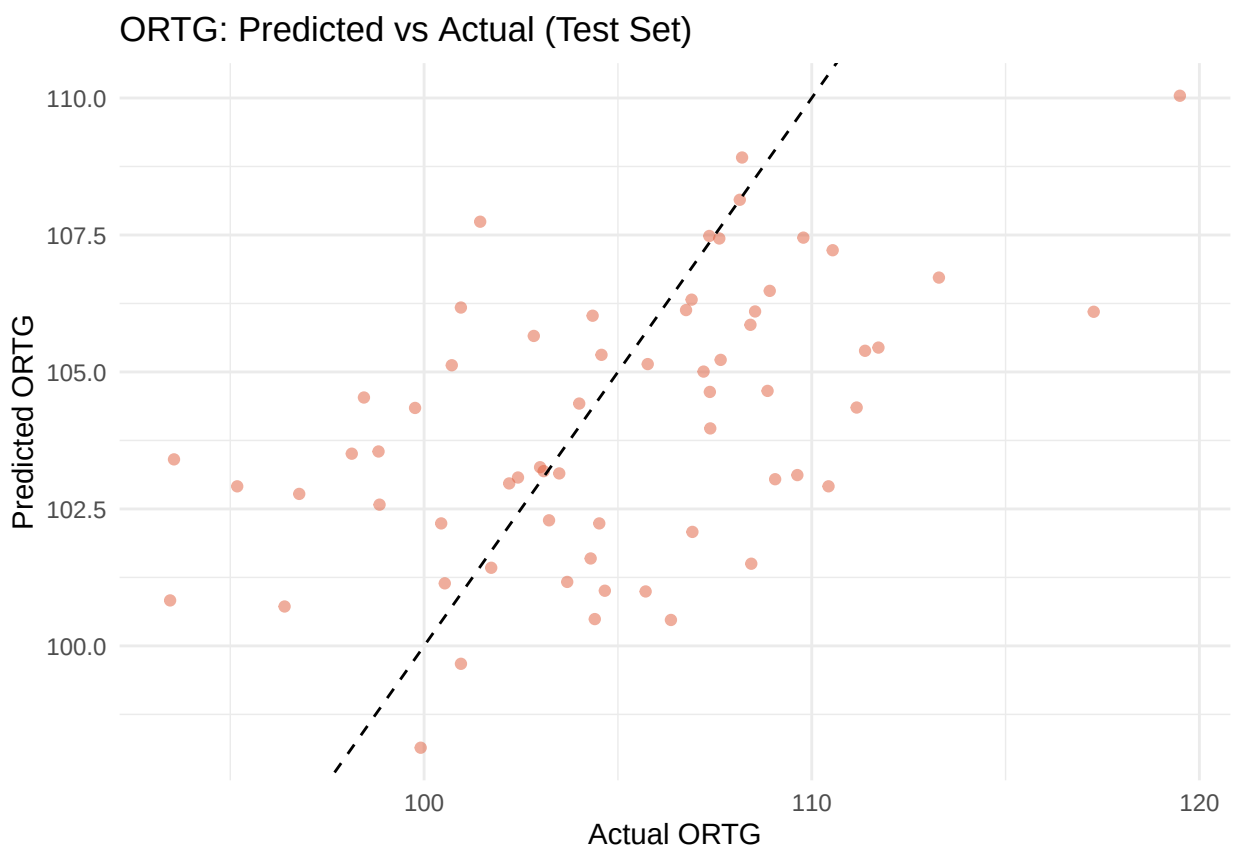
```r
final_v3 <- proj_2025 %>%
  mutate(proj_netrtg_2028 = proj_ortg_2028 - proj_drtg_2028) %>%
  select(player_name, team, age,
         proj_ortg_2028, proj_drtg_2028, proj_netrtg_2028) %>%
  arrange(desc(proj_netrtg_2028)) %>%
  mutate(Rank = row_number())

print(final_v3, n = 30)
```

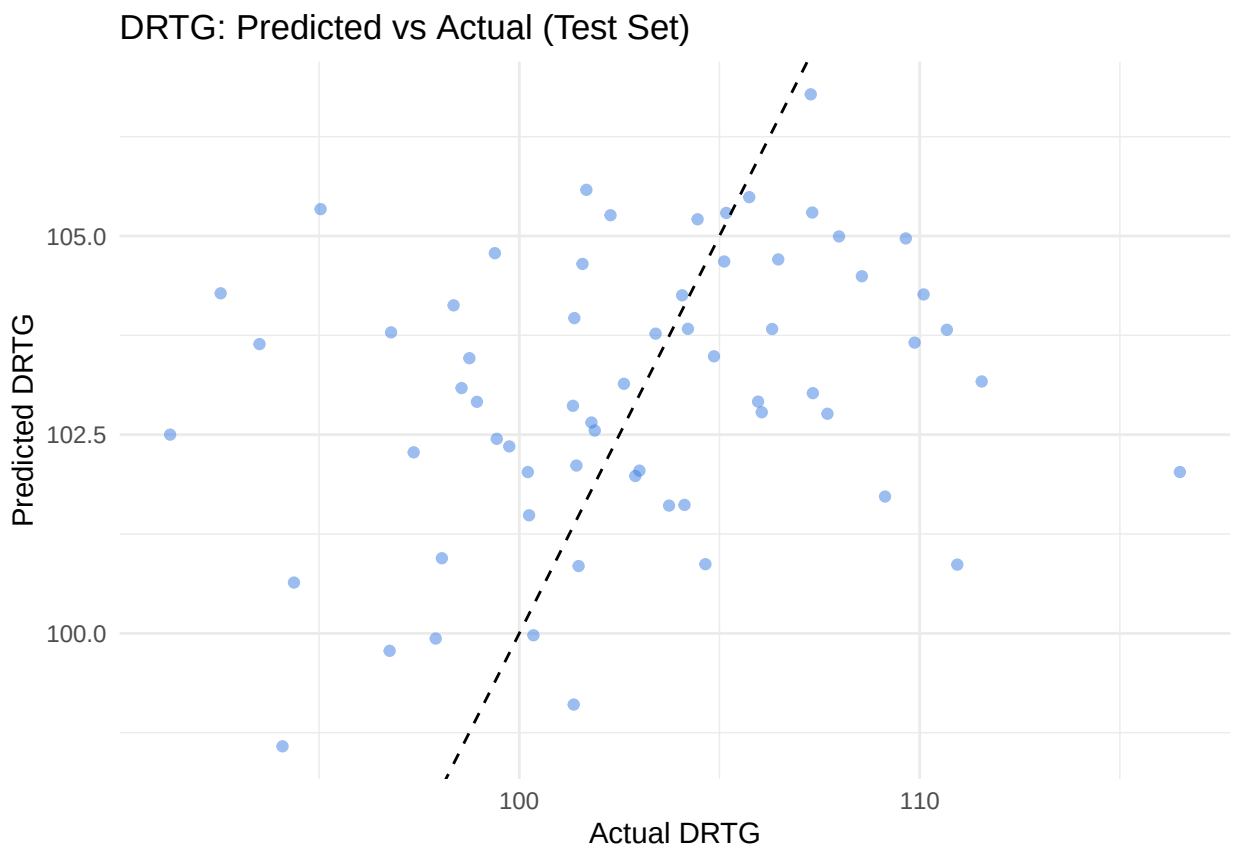
```
## # A tibble: 130 x 7
##    player_name  team    age proj_ortg_2028 proj_drtg_2028 proj_netrtg_2028  Rank
##    <chr>        <chr> <dbl>          <dbl>          <dbl>             <dbl> <int>
##  1 Napheesa Co~ MIN      32           110.           99.5             10.6      1
##  2 A'ja Wilson  LVA      32           110.           99.0             10.5      2
##  3 Caitlin Cla~ IND      27           110.          102.               8.01     3
##  4 Alyssa Thom~ PHX      36           110.          102.               7.82     4
##  5 Breanna Ste~ NYL      34           108.          100.               7.24     5
##  6 Rhyne Howard ATL      28           108.          101.               6.66     6
##  7 Jackie Young LVA      31           109.          102.               6.24     7
##  8 Alanna Smith MIN      32           107.          100.               6.17     8
##  9 Jordin Cana~ ATL      33           107.          101.               6.16     9
## 10 Courtney Wi~ MIN      34           108.          102.               6.11    10
## 11 Skylar Digg~ SEA      38           108.          102.               5.66    11
## 12 Sabrina Ion~ NYL      31           108.          103.               5.61    12
## 13 Allisha Gray ATL      34           108.          103.               5.41    13
## 14 Aliyah Bost~ IND      27           107.          102.               5.21    14
## 15 Chelsea Gray LVA      36           108.          103.               5.14    15
## 16 Kayla McBri~ MIN      36           107.          103.               4.97    16
## 17 Brionna Jon~ ATL      33           107.          102.               4.90    17
## 18 Gabby Willi~ SEA      32           105.          101.               4.63    18
## 19 Dearica Ham~ LAS      35           108.          104.               4.31    19
## 20 Jonquel Jon~ NYL      35           107.          103.               4.19    20
## 21 Kelsey Mitc~ IND      33           108.          104.               4.12    21
## 22 Emma Meesse~ NYL      35           107.          103.               4.12    22
## 23 Veronica Bu~ GSV      28           106.          102.               4.08    23
## 24 Nneka Ogwum~ SEA      38           106.          103.               3.89    24
## 25 Paige Bueck~ DAL      27           108.          104.               3.82    25
## 26 Kelsey Plum  LAS      34           109.          105.               3.63    26
## 27 Natasha Clo~ NYL      37           106.          103.               3.51    27
## 28 Satou Sabal~ PHX      30           106.          103.               3.50    28
## 29 Naz Hillmon  ATL      28           106.          103.               3.34    29
## 30 Natasha Mack PHX      31           104.          101.               3.16    30
## # i 100 more rows
```

Step 7b: Predicted vs. Actual Plots

```r
ggplot(test_set, aes(x = OnOffRtg, y = pred_ortg)) +
  geom_point(alpha = 0.5, color = "#e05c3a") +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(title = "ORTG: Predicted vs Actual (Test Set)",
       x = "Actual ORTG", y = "Predicted ORTG") +
  theme_minimal()
```




```r
ggplot(test_set, aes(x = OnDefRtg, y = pred_drtg)) +
  geom_point(alpha = 0.5, color = "#3a7de0") +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(title = "DRTG: Predicted vs Actual (Test Set)",
       x = "Actual DRTG", y = "Predicted DRTG") +
  theme_minimal()
```



Step 9a: Final Rankings Table

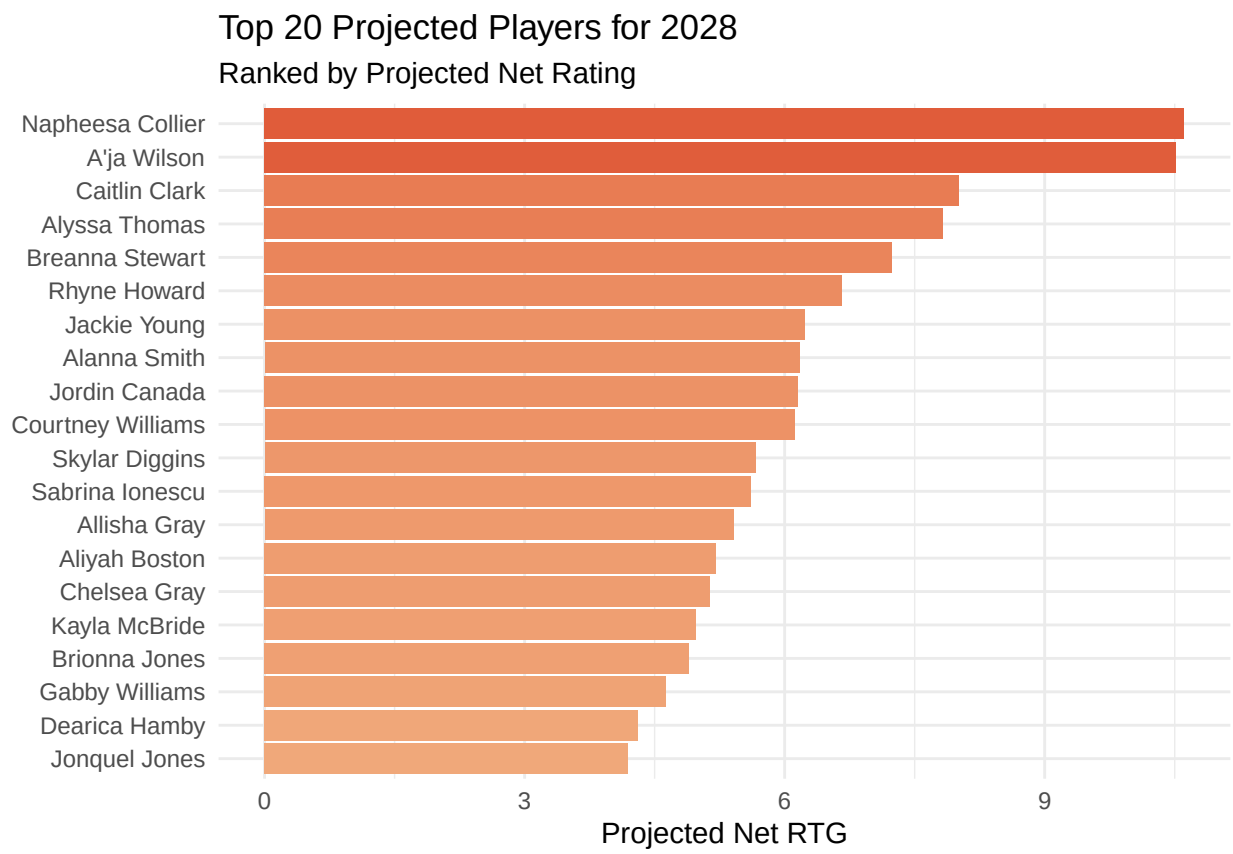
```
final_v3 %>%
  select(Rank,
         Player  = player_name,
         Team    = team,
         Age     = age,
         `Proj ORTG` = proj_ortg_2028,
         `Proj DRTG` = proj_drtg_2028,
         `Net RTG`   = proj_netrtg_2028) %>%
  mutate(across(where(is.numeric), ~ round(., 1))) %>%
  kbl(align = c("c","l","c","c","c","c","c"),
      caption = "2028 WNBA Player Projections - Ranked by Net Rating") %>%
  kable_styling(bootstrap_options = c("striped","hover","condensed"),
                full_width = TRUE,
                font_size = 10) %>%
  row_spec(0, bold = TRUE)
```

Step 9b: Bar Chart

Table 1: 2028 WNBA Player Projections — Ranked by Net Rating

Rank	Player	Team	Age	Proj ORTG	Proj DRTG	Net RTG
1	Napheesa Collier	MIN	32	110.1	99.5	10.6
2	A'ja Wilson	LVA	32	109.5	99.0	10.5
3	Caitlin Clark	IND	27	109.9	101.9	8.0
4	Alyssa Thomas	PHX	36	109.7	101.9	7.8
5	Breanna Stewart	NYL	34	107.6	100.3	7.2
6	Rhyne Howard	ATL	28	108.0	101.3	6.7
7	Jackie Young	LVA	31	108.6	102.4	6.2
8	Alanna Smith	MIN	32	106.5	100.4	6.2
9	Jordin Canada	ATL	33	107.2	101.0	6.2
10	Courtney Williams	MIN	34	108.0	101.8	6.1
11	Skylar Diggins	SEA	38	107.6	101.9	5.7
12	Sabrina Ionescu	NYL	31	108.3	102.6	5.6
13	Allisha Gray	ATL	34	108.2	102.8	5.4
14	Aliyah Boston	IND	27	107.4	102.1	5.2
15	Chelsea Gray	LVA	36	107.8	102.7	5.1
16	Kayla McBride	MIN	36	107.5	102.5	5.0
17	Brionna Jones	ATL	33	106.7	101.8	4.9
18	Gabby Williams	SEA	32	105.3	100.7	4.6
19	Dearica Hamby	LAS	35	108.0	103.7	4.3
20	Jonquel Jones	NYL	35	106.8	102.6	4.2
21	Kelsey Mitchell	IND	33	108.0	103.9	4.1
22	Emma Meesseman	NYL	35	106.7	102.6	4.1
23	Veronica Burton	GSV	28	106.4	102.4	4.1
24	Nneka Ogwumike	SEA	38	106.5	102.6	3.9
25	Paige Bueckers	DAL	27	107.6	103.7	3.8
26	Kelsey Plum	LAS	34	108.7	105.1	3.6
27	Natasha Cloud	NYL	37	106.3	102.8	3.5
28	Satou Sabally	PHX	30	106.1	102.6	3.5
29	Naz Hillmon	ATL	28	106.2	102.9	3.3
30	Natasha Mack	PHX	31	104.1	100.9	3.2
31	Natisha Hiedeman	MIN	32	106.0	102.9	3.1
32	Natasha Howard	IND	37	105.6	102.8	2.8
33	Ezi Magbegor	SEA	29	103.4	100.6	2.8
34	Aari McDonald	IND	30	106.0	103.5	2.4

```r
final_v3 %>%
  slice_head(n = 20) %>%
  ggplot(aes(x = reorder(player_name, proj_netrtg_2028),
             y = proj_netrtg_2028,
             fill = proj_netrtg_2028)) +
  geom_col() +
  coord_flip() +
  scale_fill_gradient(low = "#f0a87a", high = "#e05c3a") +
  labs(title = "Top 20 Projected Players for 2028",
       subtitle = "Ranked by Projected Net Rating",
       x = NULL, y = "Projected Net RTG") +
  theme_minimal() +
  theme(legend.position = "none")
```



Step 10: Export Files

```r
write.csv(train_set,
          '/Users/shashwatmishra/Downloads/WNBA_train_set_v2.csv',
          row.names = FALSE)

write.csv(test_set,
          '/Users/shashwatmishra/Downloads/WNBA_test_set_v2.csv',
          row.names = FALSE)
```

```r
write.csv(final_v3 %>%
            select(Rank, player_name, team, age,
                   proj_ortg_2028, proj_drtg_2028, proj_netrtg_2028),
          '/Users/shashwatmishra/Downloads/WNBA_2028_Projections_Final_V3.csv',
          row.names = FALSE)

cat("Exported train set:", nrow(train_set), "rows\n")
```

```
## Exported train set: 250 rows
```

```r
cat("Exported test set:", nrow(test_set), "rows\n")
```

```
## Exported test set: 63 rows
```

```r
cat("Exported projections:", nrow(final_v3), "players\n")
```

```
## Exported projections: 130 players
```